Board C — HV switch stage

Adafruit Perma-Proto **1/4** (columns 1–15) carrying Q1 IRFP460, D9 STTH8R06 flyback, TVS1 P6KE200A, and the local `SW` / flyback loop. All copper is in the floating HV domain referenced to `−70V`. Screw terminals **T4** and **T5** wire to the DIN K1 return contacts; motor leads land on K1 COM, not on this board.

Draft. View with the screw terminals on the **bottom** edge and the IRFP460 heatsink at the **top**. Terminals are numbered **T1–T5** left to right (T1 at column 3). On the as-built unit all four Perma-Proto power rails are **unused** — HV nets use column jumpers, not the silkscreen ± buses. On-board gate clamp parts (10 kΩ, BZX85C15) and row-column routing are documented elsewhere. Pin-by-pin IC wiring is not duplicated here.

Floating HV — up to 140 VDC. Every net on Board C is hazardous relative to earth when the motor supply is energized. `−70V` is **not** `MCU_GND`. Do not bond this board to protective earth except through an analyzed heatsink isolator. Q1 tab and drain are live at `SW`. Use an isolated or differential probe referenced to `−70V`, not earth ground. Discharge the bus and verify safe voltage before service.

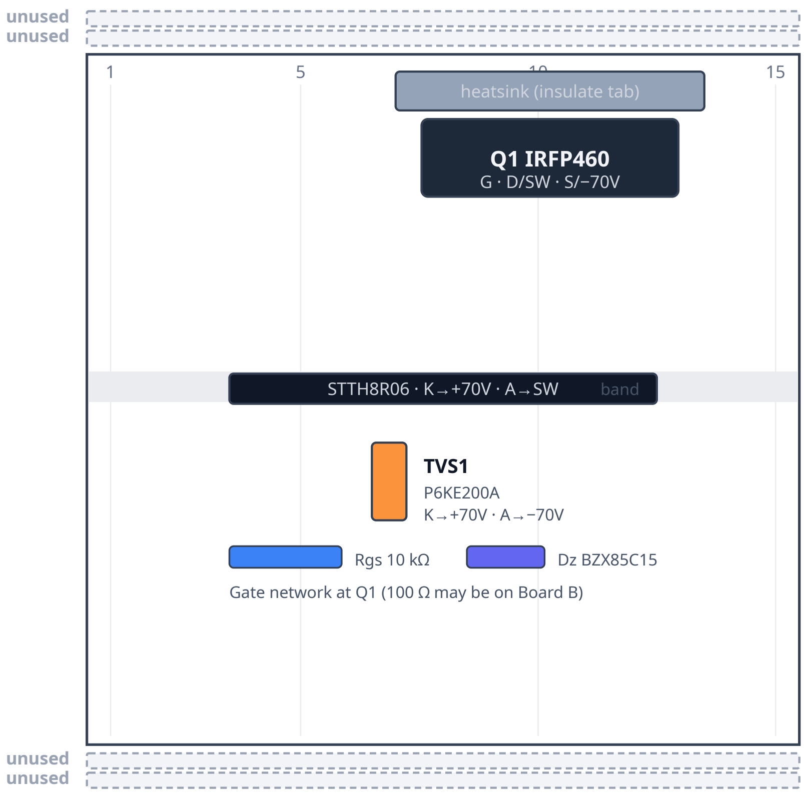
1. Placement

Mount on DIN rail with ≥20 mm clearance beside the heatsink for airflow (see DIN layout). Q1 sits under an insulated heatsink at the top of the proto. D9 spans the mid board with cathode toward the fused `+70V` side and anode to `SW`. TVS1 clamps between the on-board `+70V` and `−70V` nodes (via column wiring, not the Perma-Proto rails). Five screw terminals on the bottom edge accept field and inter-board cables; there are no terminals on the top edge.

Flyback loop constraint. When Q1 turns off, motor inductance pushes current through D9 from `SW` to `+70V`. That path — `SW` → D9 anode → D9 cathode → `+70V` — must stay physically short and entirely on Board C. Figure 1 shows this as a dashed callout box below the proto; it is a *placement rule*, not a specific jumper wire to install. Do not run the flyback loop through a long cable to Board B or the DIN rail.

Board C — placement (first draft)

Heatsink up · terminals down · floating HV only · cols 1–15



Flyback loop

Placement rule — not a wire path

SW → D9 → +70V

When Q1 turns off, motor inductance pushes current through D9.

Path: SW → D9 anode → D9 cathode → +70V.

Keep D9 physically close to the SW node and +70V landing.

Do not route this flyback loop off Board C.

BOTTOM edge · left → right

T1	T2	T3	T4	T5
+70V	−70V	GATE	NC1	NO2

Net colors

- +70V fused high rail
- SW — K1 NC1 / NO2 (T4 / T5)
- Dashed box = placement rule (not a net)
- −70V motor low / Q1 source
- GATE from Board B TC4426
- Unused Perma-Proto rail

Draft. All four Perma-Proto rails are unused on the as-built unit. T4/T5 land K1 return contacts (NC1/NO2); motor leads are on K1 COM.

As-built: T1/T2/T4/T5 leads are reinforced for motor current; T3 is gate only.

Never join MCU_GND to −70V. Q1 tab is live at SW — insulate the heatsink.

Figure 1. Grid-true placement (columns 1–15). Terminal IDs T1–T5 label each screw block left to right. Dashed gray bars are unused Perma-Proto rails. T4 and T5 are separate K1 return-contact landings (NC1 / NO2), both on the `SW` net on-

board. The full-width dashed amber box below the proto is a placement constraint for the flyback path — not a wire diagram.

2. Rails

The 1/4 Perma-Proto has four full-width buses (15 columns). On the as-built unit **all four rails are unused** — no jumpers connect them to `+70V`, `−70V`, or any other net. HV distribution uses column strips and point-to-point jumpers instead. Do not land `MCU_GND`, `+5V_EARTH`, `+5V_ISO`, or `+12V_GD` on any Board C bus.

Rail	Net	Diagram color
Upper outer −	unused (as-built; no jumper)	■ unused
Upper inner +	unused (as-built; no jumper)	■ unused
Lower inner −	unused (as-built; no jumper)	■ unused
Lower outer +	unused (as-built; no jumper)	■ unused

3. What attaches

Power

Board C receives the motor outer rails from the inter-box 140 V feed and Wago distribution (see DIN layout, row B). AnTek `+70V` is interlocked in the **PSU box** (AC-in and DC-out relays, coils on `+12V_GD`) before that feed reaches F1, the Wago star, K1, and **T1**. Run those coil leads in the **AC** inter-box passage, not beside the ±70 V PWM pair. Details: DIN enclosure — 140 V interlock relays. The fused `+70V` high rail enters at **T1** after that interlock and F1. The low outer rail `−70V` enters at **T2**. Those terminals feed D9 cathode, TVS1, and Q1 source on-board. K1 high-side contacts (NO1/NC2) take fused `+70V` from the Wago star, not through a second Board C terminal, unless your harness branches from T1.

Screw terminals

Numbering follows Figure 1: left to right with the heatsink up and terminals down. T1 is leftmost (column 3 on the as-built unit). On the as-built board, leads at the motor-current terminals (**T1**, **T2**, **T4**, **T5**) are **reinforced** for the higher continuous and PWM motor current. Use heavy-gauge, HV-rated wire on those paths; keep **T3** (gate) as a low-current signal lead. See DIN layout for 140 V path gauge guidance (18 AWG minimum; 16 AWG preferred for runs >1 m).

Bottom edge — T1–T5

Terminal	Label	Net	Attaches
T1	**+70V**	`+70V`	**Power in.** After PSU-box DC interlock, then F1 / Wago → D9 K, TVS1 K (board-internal)
T2	**−70V**	`−70V`	**Power in.** Motor low outer rail → Q1 source, TVS1 A, Rgs/Dz return (board-internal)
T3	**GATE**	gate	Board B TC4426 OUT B (100 Ω may be on B or at Q1; see Motor_Driver_Wiring.md)
T4	**NC1**	`SW`	K1 pin 12 (NC1) — return contact → Q1 drain / D9 anode on-board
T5	**NO2**	`SW`	K1 pin 24 (NO2) — return contact → same `SW` node as T4 on-board

On-board only (not on terminals)

Ref	Part	Notes
Q1	IRFP460NPBF	Low-side switch; tab = drain = `SW`
D9	STTH8R06D	Motor PWM flyback; cathode → `+70V`, anode → `SW`
TVS1	P6KE200A	Across `+70V` and `−70V` on-board (column wiring; rails unused)
Rgs	10 kΩ	Gate-to-source at Q1
Dz	BZX85C15	15 V gate clamp at Q1

Full pin tables: Motor_Driver_Wiring.md.

Board B → Board C

Keyed two-conductor floating cable from Board B **T9** (PWM out / gate) and **T3** (`−70V` gate-driver return / Q1 source reference) to Board C **T3** (GATE) and **T2** (`−70V`). Keep this harness short; do not route the D9 flyback loop off Board C.

Board C → DIN K1 (contacts)

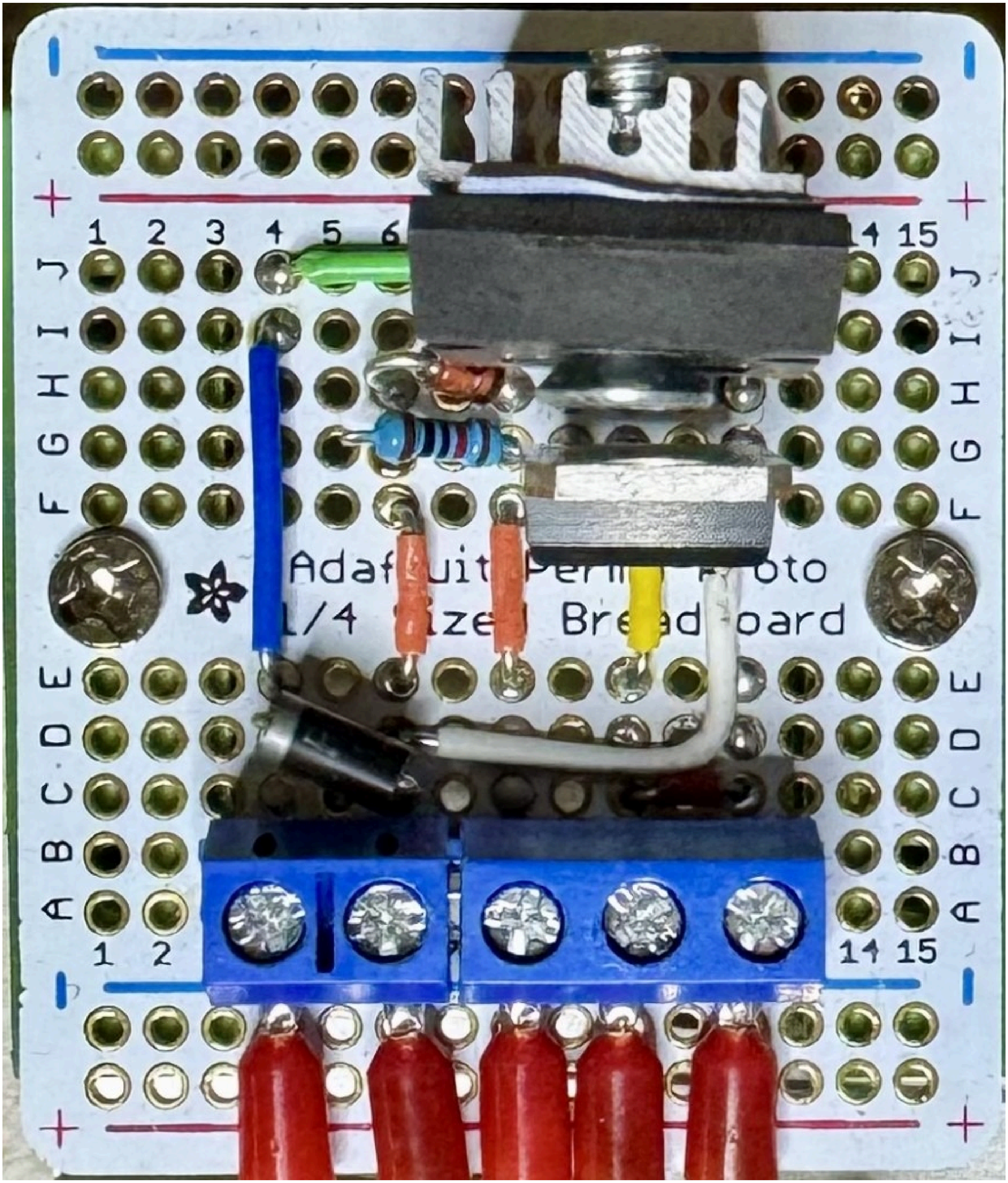
Direction switching uses the Phoenix K1 DPDT on the DIN rail. Board C lands the return-contact harness; high-side contacts and motor commons wire at the relay.

K1 pin	Net	From
NO1, NC2	`+70V` (fused)	Wago / F1 star (not a Board C terminal unless branched from T1)
NC1 (12)	`SW`	**T4**
NO2 (24)	`SW`	**T5**
COM1, COM2	`MOTOR_A`, `MOTOR_B`	Field motor harness (HV gland)

K1 coil (`+12V_GD` / Q3) is driven from Board B; see K1 floating coil option.

4. As-built photo

Line art is the labeled figure. The photo shows the unit that was built.



Wired carrier. Heatsink up, five bottom-edge terminals, columns 1–15. Red leads at T1–T2 and T4–T5 are reinforced for motor current; T3 (gate) is a low-current signal.

Related

This page is a chapter of the Wiring Guide. Electrical wiring truth for the motor stack remains the gate-driver schematic, not the placement drawing.